

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Amirtha sri N	Reg. No.:	1911260175
Date:	1.10.2026	Duration:	60 minutes

Code of Conduct

I Amirtha sri (1911260175) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing;

	strong technical justification.			report lacks organization and clarity.
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Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

Topic: AI-Based College FAQ and Student Support Chatbot

1. Understand the problem statement

Explain the problem in your own words.

In colleges, students frequently have questions about academic schedules, attendance, examinations, courses, fees, departments, events, classrooms, and college procedures. Currently, students usually depend on faculty members, administrative staff, notice boards, WhatsApp groups, websites, or office visits to obtain this information. This can be time-consuming and students may not receive immediate or consistent responses.

The proposed AI-Based College FAQ and Student Support Chatbot is a software system that provides quick answers to common student questions. The chatbot uses Artificial Intelligence and Natural Language Processing (NLP) to understand student queries and provide relevant information. It can be accessed through a web or mobile interface and can support students at any time.

The main objectives are:

- To provide instant answers to frequently asked college-related questions.
- To reduce the workload of faculty and administrative staff.
- To provide students with information through a single platform.
- To understand student questions using AI and Natural Language Processing.
- To provide information about attendance, timetable, examinations, courses, fees, departments, and events.
- To improve accessibility to college information.
- To reduce repeated enquiries to college staff.
- To provide personalized student support when required

Scope of the problem:

The scope of the chatbot covers common student support activities such as college FAQs, academic information, examination details, attendance-related information, department details, timetable information, campus facilities, events, and basic administrative queries. The system can be extended in the future to support multiple languages, voice-based interaction, personalized notifications, student portals, and integration with existing college management systems.

2. Identify Key Stakeholders and Challenges

Identify all the stakeholders involved in the system

The major stakeholders are:

- 1. Students – Use the chatbot to ask questions and obtain college information.**
- 2. Faculty Members – Provide academic information and support students.**

- 3. Administrative Staff – Manage administrative information and student enquiries.**
- 4. College Management – Monitors the system and ensures proper implementation.**
- 5. System Administrator – Manages users, chatbot data, security, and system configuration.**
- 6. IT/Development Team – Develops, maintains, tests, and updates the chatbot.**
- 7. Examination Department – Provides examination-related information.**
- 8. Admission/Accounts Department – Provides admission and fee-related information.**

Roles and responsibilities

Students enter questions and receive answers from the chatbot.

- 1. Faculty members provide accurate academic information and help update subject-related FAQs.**
- 2. Administrative staff maintain information related to fees, procedures, documents, and college services.**
- 3. College management supervises the system and ensures that the chatbot supports institutional requirements.**
- 4. System administrators manage access permissions, chatbot knowledge, system security, and updates.**
- 5. Development team develops the chatbot, integrates AI/NLP services, performs testing, and fixes technical issues.**

Major challenges

The stakeholders may face several challenges:

- 1. Students may not know where to find correct information.**
- 2. Faculty and administrative staff receive repeated questions.**
- 3. Information may be distributed across different platforms.**
- 4. Manual responses can take time.**
- 5. Incorrect or outdated information may confuse students.**
- 6. The chatbot may not understand unclear or complex questions.**
- 7. Student data must be protected from unauthorized access.**
- 8. Regular updating of chatbot information is required.**

3. Analyze the Current Approach

Existing system/current process

In the traditional college system, students obtain information through notice boards, college websites, WhatsApp groups, emails, faculty members, administrative offices, and direct enquiries.

For example, a student who wants to know the examination date may have to check a notice, website, WhatsApp message, or contact a faculty member. Different information sources may make it difficult to find the required information quickly.

Strengths

- **Existing methods are familiar to students and staff.**
- **Faculty members can provide personalized support.**
- **Official notices can provide reliable information.**
- **Websites and portals can store large amounts of information.**

Limitations

- **Students may need to search through multiple platforms.**
- **Responses may not be available immediately.**
- **Faculty members receive repeated questions.**
- **Information may be outdated on some platforms.**
- **Manual communication requires more time.**
- **Students may have difficulty finding the exact information they need.**

Problems or gaps

The major gap is the absence of a centralized intelligent support system that can understand student questions and provide immediate answers.

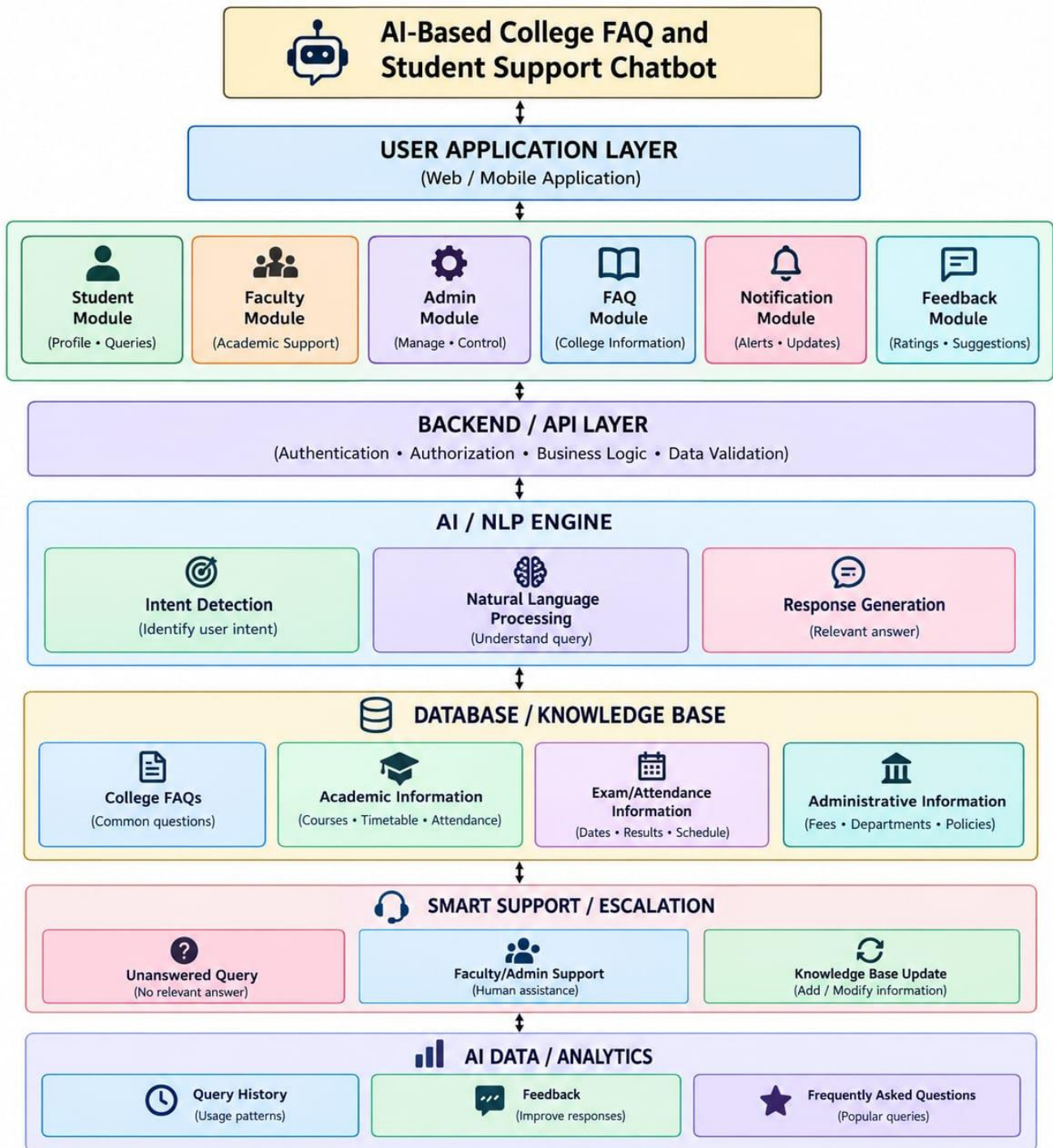
The proposed chatbot addresses this gap by providing a single conversational interface where students can ask questions in natural language and receive relevant responses.

4. Propose a Suitable Solution Using Software Engineering Principles

Design of the proposed solution

The proposed AI-Based College FAQ and Student Support Chatbot is an intelligent software platform that allows students to communicate with an AI chatbot.

The system contains a student interface, chatbot/NLP module, knowledge base, backend services, database, authentication system, and administrator dashboard.



Simple system flow

Student



Chatbot Interface



Natural Language Processing (NLP)



Query Understanding



College FAQ / Knowledge Base



Relevant Answer



Student

If the chatbot cannot answer a query:

Unanswered Query → Faculty/Admin Support → Updated Knowledge Base

Major functional modules

1. Student Registration and Login Modules

Students can securely log in and access the chatbot.

2. Chatbot Module

Students can enter questions using natural language and receive answers.

3. NLP Module

The system analyzes the student's question and identifies its meaning or intent.

4. FAQ and Knowledge Base Module

Stores frequently asked questions and verified college information.

5. Academic Information Module

Provides information related to courses, timetable, attendance, examinations, and academic activities.

6. Administrative Support Module

Provides information about fees, documents, admission procedures, departments, and college services.

7. Notification Module

Can provide important updates such as examination dates, events, deadlines, and announcements.

8. Admin Module

Administrators can add, update, or remove information from the chatbot knowledge base.

9. Feedback Module

Students can provide feedback about whether the chatbot response was useful.

Functional Requirements

- **Student registration and login.**
- **Chatbot-based question answering.**
- **Natural language query processing.**

- **FAQ search.**
- **Academic information retrieval.**
- **Examination information.**
- **Attendance-related information.**
- **Department and course information.**
- **College event information.**
- **Administrative information.**
- **Admin login and management.**
- **Knowledge-base updating.**
- **Feedback collection.**
- **Chat history management.**

Non-Functional Requirements

- **Security:** Student information must be protected using authentication and authorization.
- **Reliability:** The chatbot should provide accurate and consistent responses.
- **Scalability:** The system should support increasing numbers of students.
- **Usability:** The interface should be simple and easy to use.
- **Maintainability:** Information and chatbot modules should be easy to update.
- **Performance:** Responses should be provided within an acceptable time.
- **Availability:** The chatbot should be accessible whenever students need support.
- **Accuracy:** The system should provide verified college information.

Software Engineering Principles

- **Modularity:** The system is divided into independent modules such as chatbot, NLP, FAQ, authentication, admin, and feedback.
- **Scalability:** Cloud-based architecture can allow the system to support more students and queries as usage increases.
- **Maintainability:** The chatbot, database, and user interface can be maintained separately, making updates easier.
- **Reliability:** Verified information, validation, error handling, and regular testing can improve reliability.
- **Usability:** A simple conversational interface allows students to ask questions without navigating through many pages.
- **Security:** Authentication, authorization, secure communication, and controlled access can protect student information.

5. Justify the Chosen Methodology/Framework

Scrum Selected methodology: Agile with

Agile methodology with Scrum is suitable for developing the AI-Based College FAQ and Student Support Chatbot.

The chatbot contains several modules such as authentication, FAQ management, NLP, chatbot interaction, admin management, and notifications. These features can be developed incrementally through short development cycles called sprints.

Why Agile Scrum is suitable

In the first sprint, the team can develop the basic chatbot interface and FAQ system.

In the second sprint, NLP-based question understanding can be added.

In the third sprint, academic and administrative modules can be integrated.

Later sprints can add notifications, feedback, personalization, and advanced AI features.

Agile allows students, faculty, and administrators to provide feedback during development. This feedback can be used to improve chatbot responses and usability.

Support for development and maintenance

Scrum provides regular planning, development, testing, review, and feedback. Continuous testing can identify errors early. After deployment, new FAQs, college policies, academic information, and AI features can be added through future iterations.

6. Discuss Expected Outcomes and Limitations

Expected benefits

The proposed system is expected to:

- Provide instant student support.
- Reduce repeated enquiries to faculty and administrative staff.
- Improve access to college information.
- Provide information through a centralized platform.
- Save students time
- Improve communication between students and the institution.

- Provide 24/7 automated support.
- Reduce manual workload.
- Improve the overall student experience.

How the solution addresses challenges

The chatbot addresses fragmented information by providing a centralized platform for common college information. It reduces repeated enquiries by automatically answering frequently asked questions. NLP helps the system understand questions written in natural language.

The administrator can regularly update the knowledge base to ensure that students receive current information.

Authentication and access control help protect student-related information.

Risks, implementation challenges, and limitations

The system may face the following challenges:

- The chatbot may misunderstand complex questions.
- Incorrect information may be provided if the knowledge base is not updated.
- AI responses require regular monitoring.
- Student data must be securely protected.
- Internet connectivity is required for online access.
- Development and maintenance may require technical expertise.
- Some questions may require direct support from faculty or administrative staff.
- AI cannot completely replace human support for complex or sensitive issues.

Future enhancements

Future versions can include:

- Voice-based chatbot interaction.
- Multi-language support, including regional languages.
- Personalized answers based on student profiles.

- **Integration with college ERP/Student Portal.**
- **Automated attendance information.**
- **Examination reminders and notifications.**
- **AI-based academic assistance.**
- **Mobile application.**
- **WhatsApp or other communication-platform integration.**
- **Analytics to identify frequently asked student questions.**
- **Improved AI models for complex queries.**

Conclusion

The AI-Based College FAQ and Student Support Chatbot can provide a centralized and intelligent platform for answering common student queries. By using AI, NLP, modular design, security, scalability, and Agile Scrum methodology, the proposed system can improve student support while reducing the workload of faculty and administrative staff. Therefore, Agile Scrum supports both the initial development and long-term maintenance of the chatbot.